Invitation to Search or Purchase? Optimal Multi-attribute Advertising

Yunfei (Jesse) Yao

The Chinese University of Hong Kong

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Motivation

- Consumers often evaluate different attributes of a product before making a purchasing decision
- Learning requires time and effort
- Firms want to encourage consumers to purchase or continue seeking information rather than abandoning search
- May provide information through advertising

Motivation

- Advertising can raise consumers' interest
  - Tesla's ad highlighting its interior design featuring a large screen for a car buyer who values electronic entertainment systems
- Advertising may also create a negative impression
  - The same ad for a different buyer who prefers the mechanical feel of a car

Motivation

- Limited bandwidth of ads (Shapiro 2006, Bhardwaj et al. 2008)
  - A firm cannot communicate all available information
  - Needs to decide on which attributes to emphasize

Motivation

- The consumer can search for additional information after seeing an ad
- Could focus on selected attributes
  - A consumer might not investigate the design of a Tesla due to its well-known styling
- Or learn about all attributes
  - A consumer might investigate all attributes of Faraday Future (significant uncertainty about all aspects)
- A firm's advertising strategy affects the consumer's search problem.

Research Questions

- Whether the firm wants to advertise or not?
- What is the optimal advertising content - which attribute should the firm advertise?
- What is the role of advertising? Does the firm want to invite the consumer to purchase the product directly or to search for additional information?
- What should be the KPI for an advertising campaign?

Contribution

- Examining the choice of specific and feasible advertising content while allowing for the possibility of consumer search

Results Preview

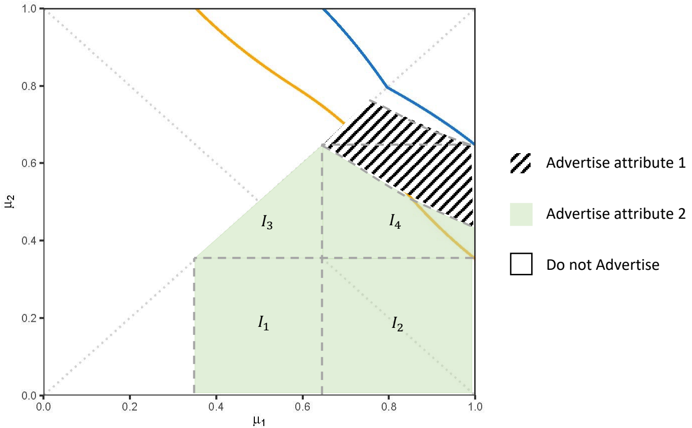


Figure: Optimal Advertising Strategy

Results Preview

Extreme prior beliefs about both attributes:

- The firm does not advertise
- No advertising: invitation to search (purchase) when the belief is high (very high);

Results Preview

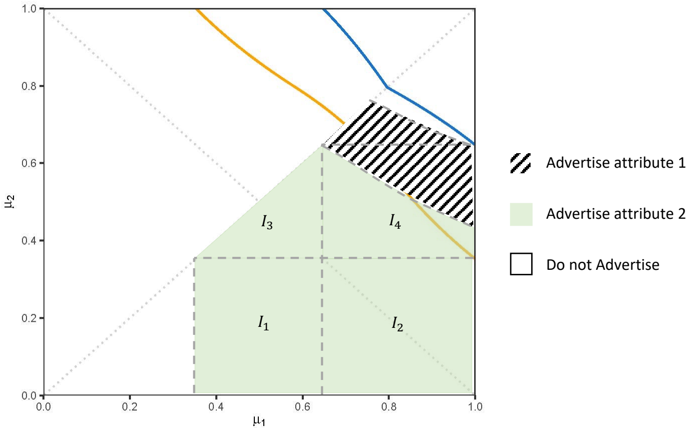


Figure: Optimal Advertising Strategy

Results Preview

Extreme prior beliefs about both attributes:

- The firm does not advertise

- No advertising: invitation to search (purchase) when the belief is high (very high);

Otherwise:

- Advertise the better attribute if the consumer is optimistic enough about the worse attribute;

  Advertise the worse attribute if the consumer is less optimistic

- Role of advertising: non-monotonic in the belief

  Reason: different binding incentives of the firm in different belief regions.

Model

- A firm and a consumer, both risk-neutral
- The consumer decides whether to purchase a product
- Two attributes with independent values (horizontal match)

  $U = U_1 + U_2,\ U_i \in \{0, 1\}$

  Consumer's prior belief that attribute i is good: $\mu_i(0)$
- Price $p \geq 3/2$: exogenous (so the consumer will not buy if one attribute is known to be bad)
- Time is continuous, $t \in [0, +\infty)$

Model

- The firm can reveal the value of one attribute by informative advertising (limited bandwidth)
- Firm advertises attribute $i \in \{1, 2\}$
  $$\Rightarrow U_i = \begin{cases} 1, \textit{with probability } \mu_i \\ 0, \textit{with probability } 1 - \mu_i \end{cases}$$
- Full disclosure: sharpest result (a noisy signal extension)

Model

- Consumers can learn more about the attributes
- If the firm advertises one attribute, they may search for information about the other attribute
- If the firm does not advertise, they may search for information about either attribute
- Flow search cost: c
  Noisy signal: $dS_i(t) = U_i dT_i(t) + \sigma dW_i(T_i(t))$
- Belief evolution:
  $d\mu_i(t) = \frac{1}{\sigma^2}\mu_i(t)[1 - \mu_i(t)]\{dS_i(t) - \mathbb{E}[U_i|\mathcal{F}_t]dT_i(t)\}$

Consumer's Search Strategy if the Firm Advertises One Attribute

- Firm advertises attribute $i \in \{1, 2\}$
  $\Rightarrow U_i = \begin{cases} 1, \textit{with probability } \mu_i \\ 0, \textit{with probability } 1 - \mu_i \end{cases}$
- Consumer may only search for information about the other attribute j: a one-dimensional search problem
- Equivalent to considering a single-attribute product whose value is U_j and whose price is $p' := p - U_i$.
- Consumer will quit if attribute i is bad
- Focus on the case in which attribute i is good

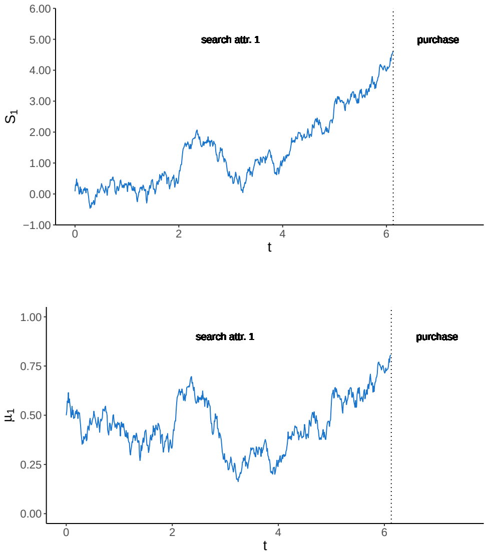
Consumer's Search Strategy if the Firm Advertises One Attribute

- Ke and Villas-Boas (2019) has characterized the optimal search strategy:

  The consumer

- searches for more information if $\mu_j \in (\underline{\mu}_j, \bar{\mu}_j)$,
- purchases the product if $\mu_j \geq \bar{\mu}_j$,
- and quits if $\mu_j \leq \underline{\mu}_j$.

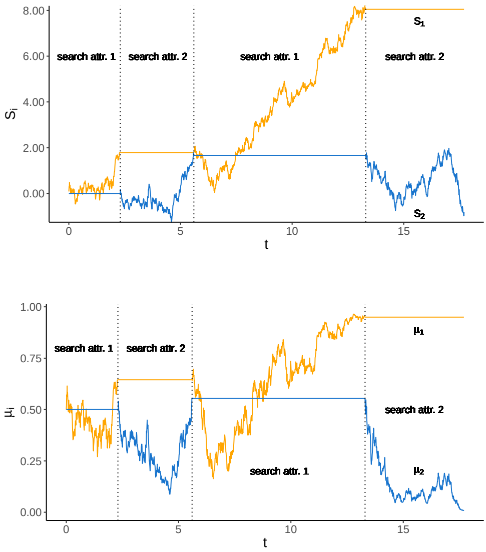
Sample Path of Signals and Belief Evolution When Both Attributes Are Good and the Firm Advertises Attribute 2



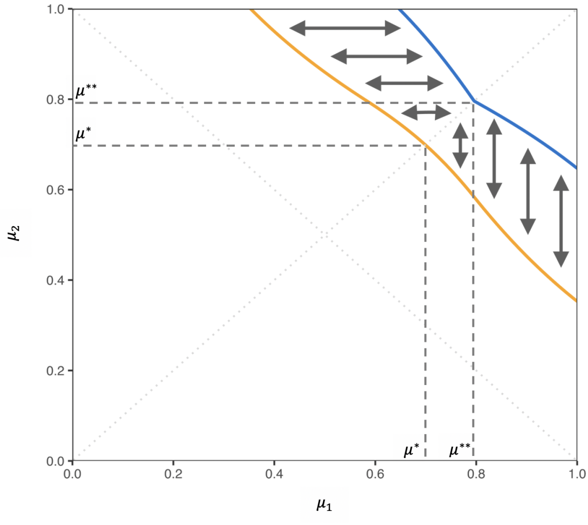
Consumer's Search Strategy if the Firm Does Not Advertise

- Consumers face a two-dimensional search problem
- They have uncertainty about both attributes
- Can search for information about either attribute

Sample Path of Signals and Belief Evolution When $U_1 = 1, U_2 = 0$, and the Firm Does Not Advertise



Consumer's Search Strategy if the Firm Does Not Advertise (Low Search Cost c)



purchasing boundary $(\mu_1, \bar{\mu}(\mu_1))$

quitting boundary $(\mu_1, \underline{\mu}(\mu_1))$

, when $\mu_1 \geq \mu_2$

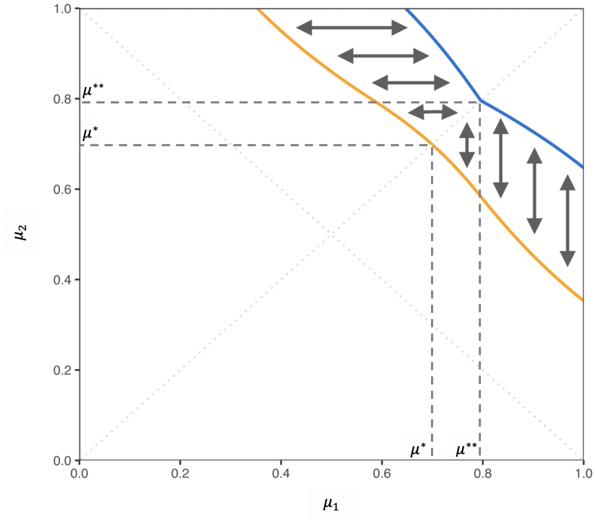
Firms' Advertising Strategy: Advertising One Attribute

Proposition

Suppose the firm advertises attribute one. The probability that the consumer purchases the product is:

$$
\begin{aligned}
& P_1(\mu_1, \mu_2) \\
:= \ & \mathbb{P}[\textit{purchasing} | \textit{firm advertises attribute one, prior belief } (\mu_1, \mu_2)] \\
= \ & \begin{cases} \mu_1, & \textit{if } \mu_2 \geq \bar{\mu}(1) \\ \mu_1 \cdot \frac{\mu_2 - \underline{\mu}(1)}{\bar{\mu}(1) - \underline{\mu}(1)}, & \textit{if } \mu_2 \in [\underline{\mu}(1), \bar{\mu}(1)] \\ 0, & \textit{if } \mu_2 \leq \underline{\mu}(1) \end{cases}
\end{aligned}
$$

Firms' Advertising Strategy: Advertising One Attribute



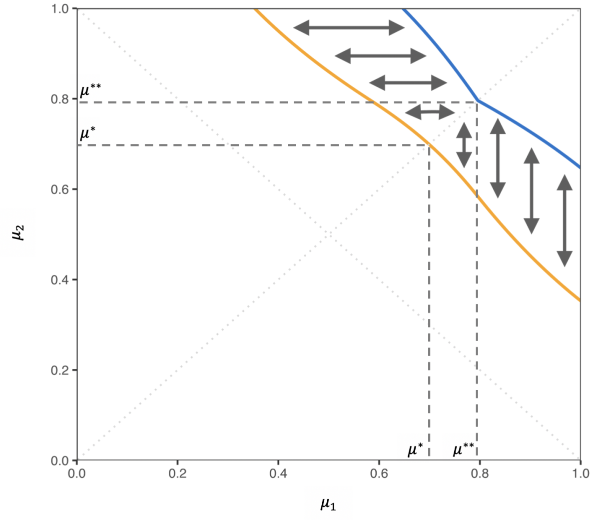
Firms' Advertising Strategy: Advertising One Attribute

Corollary

Suppose the firm advertises attribute two. The probability that the consumer purchases the product is:

$$
\begin{aligned}
& P_2(\mu_1, \mu_2) \\
:= \; & \mathbb{P}[\textit{purchasing} | \textit{firm advertises attribute two, prior belief } (\mu_1, \mu_2)] \\
= \; & \begin{cases} \mu_2, & \textit{if } \mu_1 \geq \bar{\mu}(1) \\ \mu_2 \cdot \frac{\mu_1 - \underline{\mu}(1)}{\bar{\mu}(1) - \underline{\mu}(1)}, & \textit{if } \mu_1 \in [\underline{\mu}(1), \bar{\mu}(1)] \\ 0, & \textit{if } \mu_1 \leq \underline{\mu}(1) \end{cases}
\end{aligned}
$$

Firms' Advertising Strategy: Not Advertising

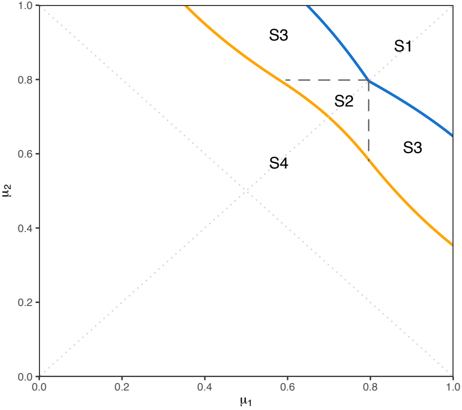


Firms' Advertising Strategy: Not Advertising

$$\mathbb{P}[\text{purchasing}|(\mu_1, \mu_2)]$$

$$= \begin{cases} 1, & \text{in } S_1 \\ \frac{\mu_2 - \underline{\mu}(\mu_1)}{\mu_1 - \underline{\mu}(\mu_1)} \; e^{-\int_{\mu_1}^{\mu^{**}} \frac{2}{x - \underline{\mu}(x)} dx}, & \text{in } S_2 \\ \frac{\mu_2 - \underline{\mu}(\mu_1)}{\bar{\mu}(\mu_1) - \underline{\mu}(\mu_1)}, & \text{in } S_3 \\ 0, & \text{in } S_4 \end{cases}$$

, when $\mu_1 \geq \mu_2$

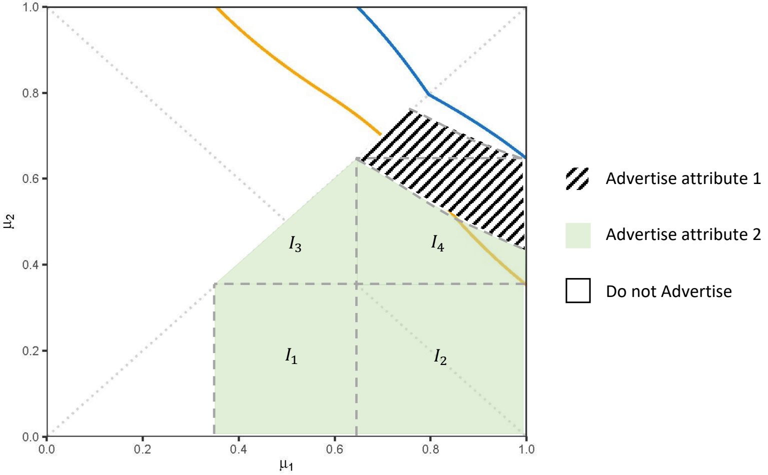


Optimal Advertising Strategy (when $\mu_1 \geq \mu_2$)

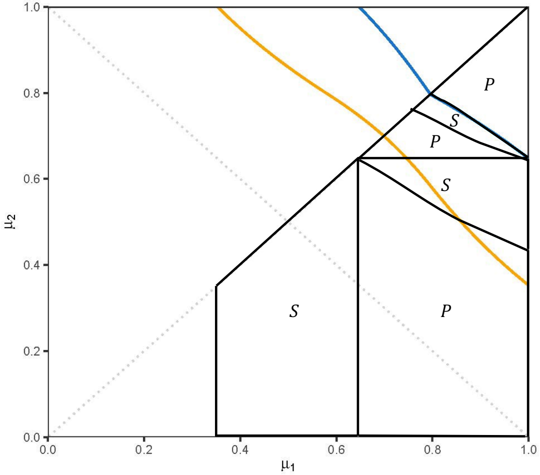
Two trade-offs:

- Whether (or not) to advertise
  - Pro: a higher likelihood of purchase if the advertised attribute is good
  - Con: no chance of sales if the advertised attribute is bad
- Whether to advertise the better attribute (or worse attribute)
  - Pro: a higher chance of keeping the consumer interested
  - Con: lower conditional evaluation and purchasing probability

Optimal Advertising Strategy



Consumer Behavior Induced by the Advertising Strategy



S Invitation to search

P Invitation to purchase

The Firm's Binding incentive

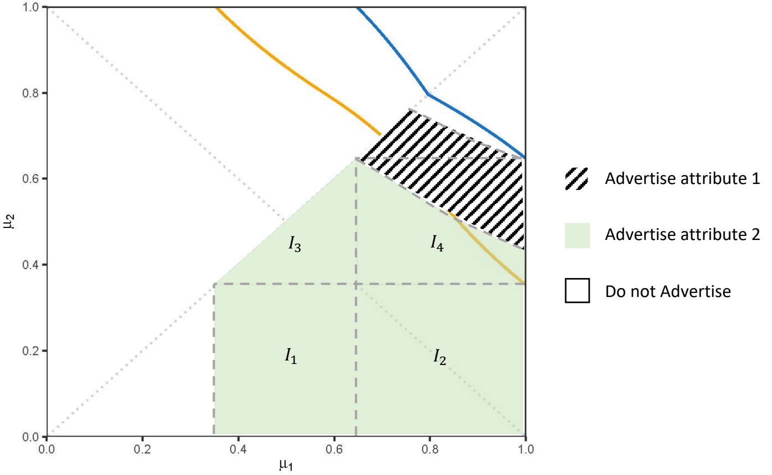
- Consumer is very pessimistic about an attribute:

  to convince the consumer to consider the product

- Less extreme belief:

  to avoid or reduce the amunt of **wasted belief**

Optimal Advertising Strategy (Blank Region)



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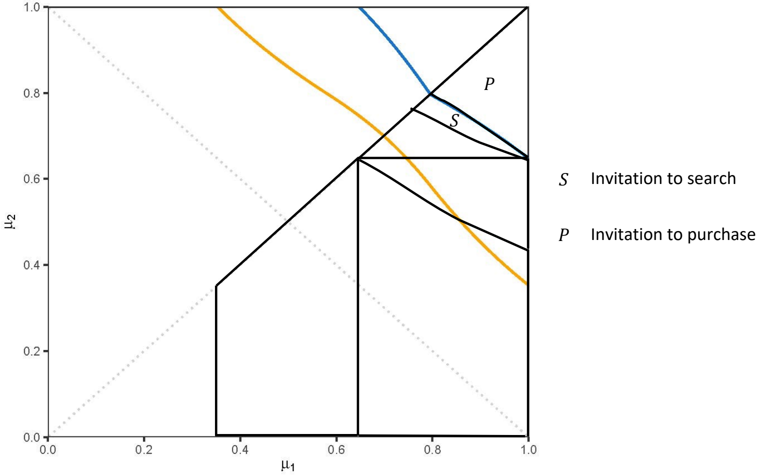
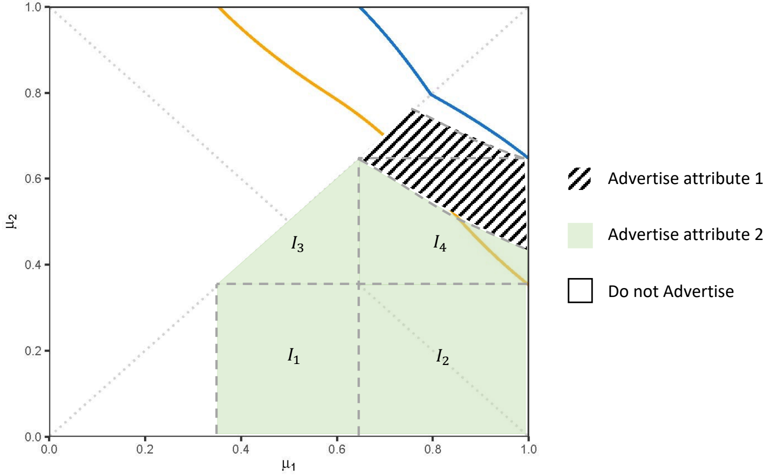
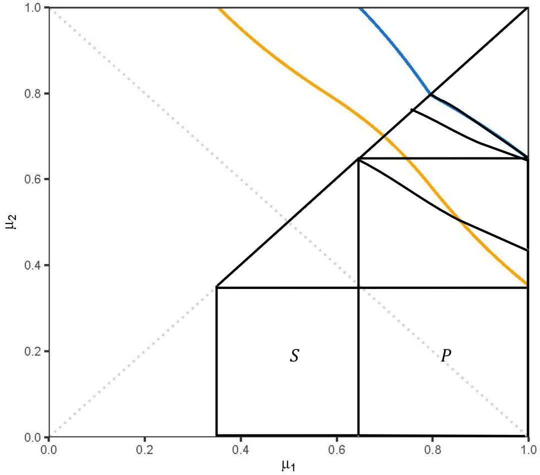


Optimal Advertising Strategy (Region I_1 & I_2)



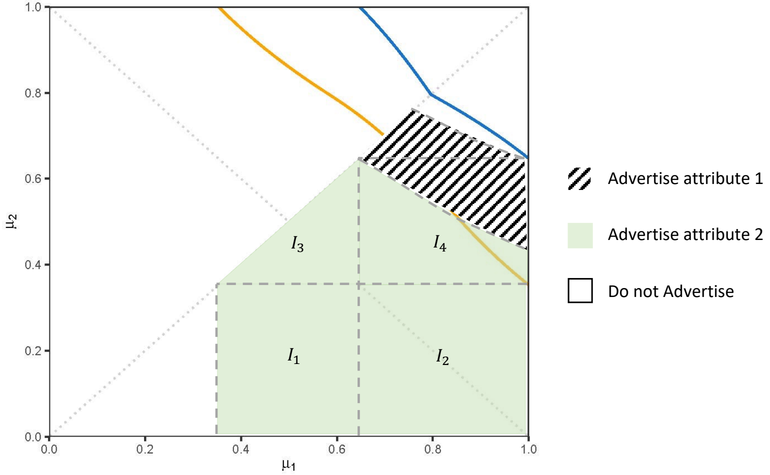
Optimal Advertising Strategy (Region I_1 & I_2)



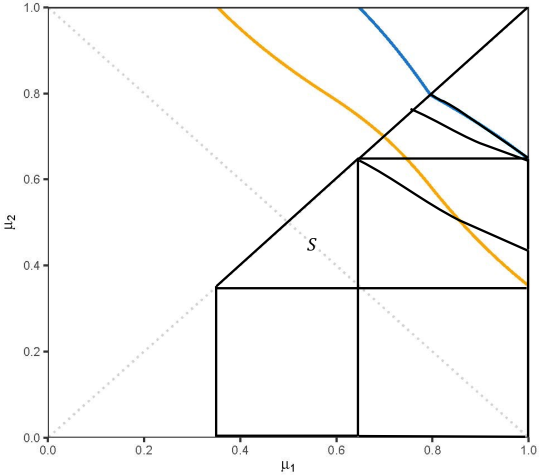
S Invitation to search

P Invitation to purchase

Optimal Advertising Strategy (Region I_3)



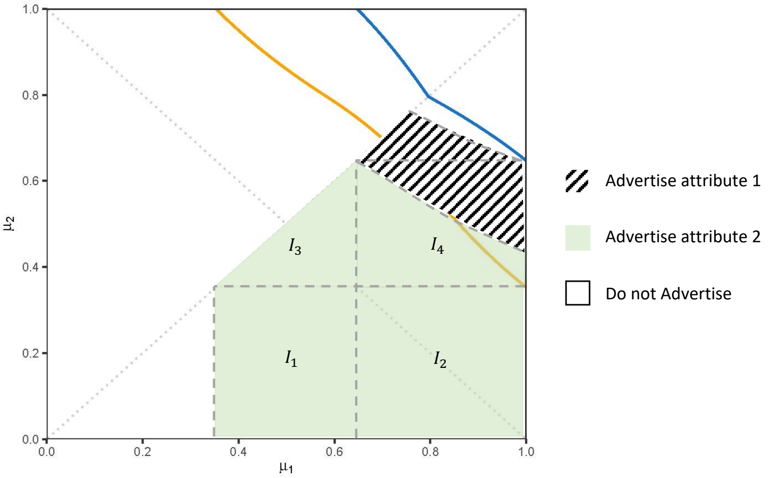
Optimal Advertising Strategy (Region I_3)



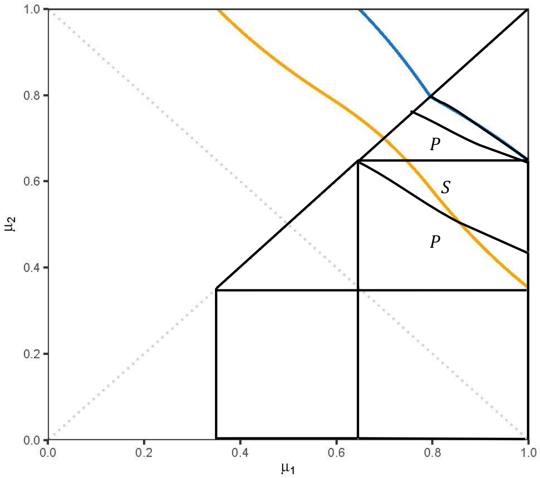
S Invitation to search

P Invitation to purchase

Optimal Advertising Strategy (Region I_4 and the Diagonal Striped Black Region)



Optimal Advertising Strategy (Region I_4 and the Diagonal Striped Black Region)



S Invitation to search

P Invitation to purchase

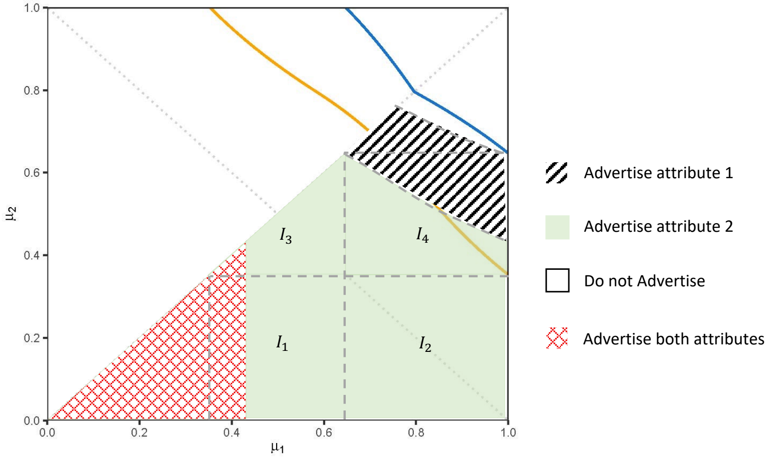
Extension: Relaxing Limited Bandwidth (Allowing Advertising Both Attributes)

- The same setup as in the main model, except that:
- The firm can also advertise both attributes

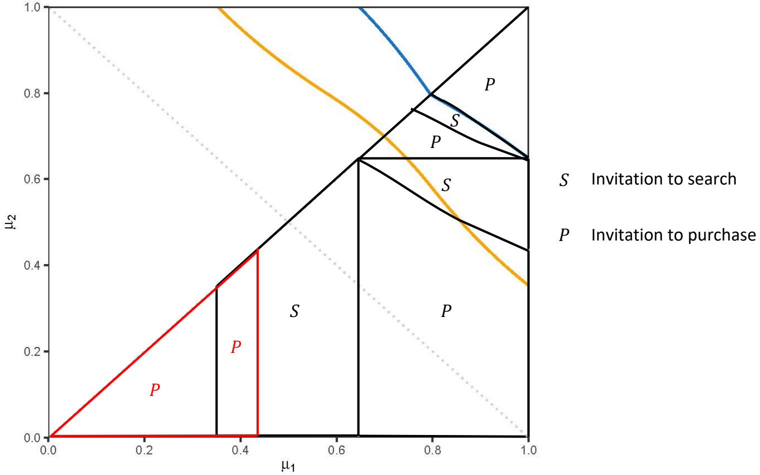
  Uncertainty about both attributes are resolved

  $\Rightarrow$ the consumer purchases iff both attributes are good

Extension: Relaxing Limited Bandwidth



Extension: Relaxing Limited Bandwidth



Extension: Noisy Signal from Advertising

- An ad may not resolve all the uncertainty about one attribute
- The same setup as in the main model, except that:
- A noisy signal $s_i \in \{g, b\}$ about attribute i
  - Signal realization matches the true state with prob. $\gamma > 1/2$
    $\mathbb{P}(s_i = g | \text{attri. } i \text{ is good}) = \mathbb{P}(s_i = b | \text{attri. } i \text{ is bad}) = \gamma$
  - $\gamma = 1$: main model
  - $\gamma < 1$: uncertainty about the value of the advertised attribute

Extension: Noisy Signal from Advertising

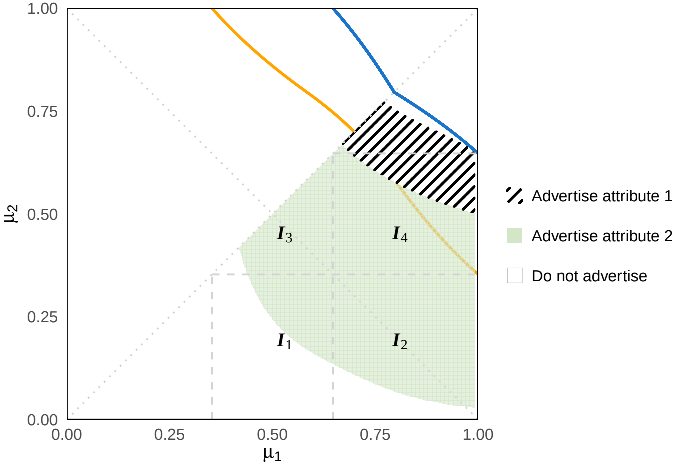


Figure: More Precise Signal, $\gamma = .95$

Extension: Noisy Signal from Advertising

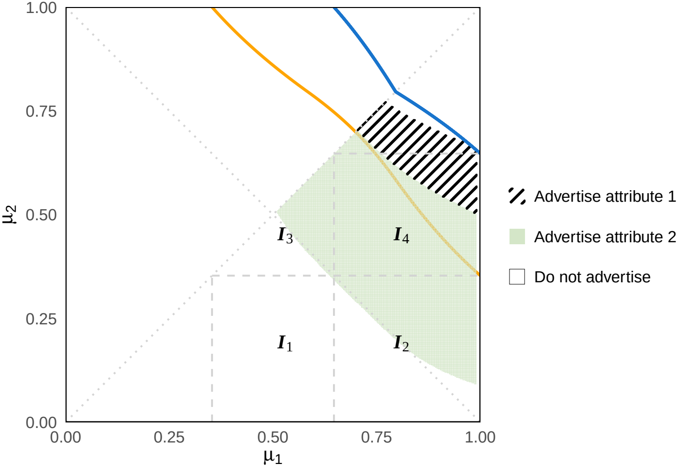
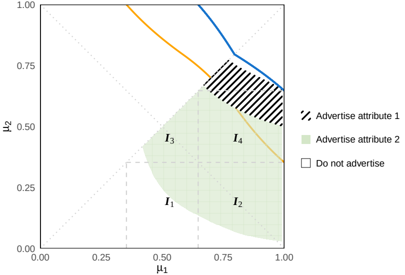


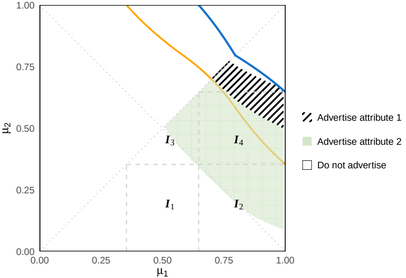
Figure: Noisier Signal, $\gamma = .85$

Extension: Noisy Signal from Advertising

More precise signal
$\gamma = .95$:



Noisier signal
$\gamma = .85$:



Thank you!